

Foundation Tier

Externally assessed

Description

The Pearson Edexcel International GCSE in Mathematics (Specification A) requires students to demonstrate application and understanding of the following.

Number

- Use numerical skills in a purely mathematical way and in real-life situations.

Algebra

- Use letters as equivalent to numbers and as variables.
- Understand the distinction between expressions, equations and formulae.
- Use algebra to set up and solve problems.
- Demonstrate manipulative skills.
- Construct and use graphs.

Geometry

- Use properties of angles.
- Understand a range of transformations.
- Work within the metric system.
- Understand ideas of space and shape.
- Use ruler, compasses and protractor appropriately.

Statistics

- Understand basic ideas of statistical averages.
- Use a range of statistical techniques.
- Use basic ideas of probability.

Students should be able to demonstrate **problem-solving skills** by translating problems in mathematical or non-mathematical contexts into a process or a series of mathematical processes.

Students should be able to demonstrate **mathematical reasoning skills** by:

Assessment information

Each paper is assessed through a 2-hour examination set and marked by Pearson.

The total number of marks for each paper is 100.

Each paper will assess the full range of targeted grades at Foundation Tier (5–1).

Each paper will have approximately equal marks available for each of the targeted grades.

There will be approximately 40% of questions targeted at grades 5 and 4, across papers 1F and 1H to aid standardisation and comparability of award between tiers.

Diagrams will not necessarily be drawn to scale and measurements should not be taken from diagrams unless instructions to this effect are given.

Each student may be required to use mathematical instruments, e.g. pair of compasses, ruler, protractor.

A Foundation Tier formulae sheet (*Appendix 4*) will be included in the written examinations.

Tracing paper may be used in the examinations.

A calculator may be used in the examinations (please see *page 42* for further information).

Questions will be set in SI units (international system of units).

AO1 Numbers and algebra

1 Numbers and the number system

	Students should be taught to:	Notes
1.1 Integers	A understand and use integers (positive, negative and zero)	
	B understand place value	
	C use directed numbers in practical situations	e.g. temperatures
	D order integers	
	E use the four rules of addition, subtraction, multiplication and division	
	F use brackets and the hierarchy of operations	
	G use the terms 'odd', 'even', 'prime numbers', 'factors' and 'multiples'	
	H identify prime factors, common factors and common multiples	
1.2 Fractions	A understand and use equivalent fractions, simplifying a fraction by cancelling common factors	$\frac{8}{60} = \frac{2}{15}$ in its simplest form (lowest terms)
	B understand and use mixed numbers and vulgar fractions	
	C identify common denominators	
	D order fractions and calculate a given fraction of a given quantity	
	E express a given number as a fraction of another number	
	F use common denominators to add and subtract fractions and mixed numbers	$\frac{2}{3} + \frac{5}{7}, \quad 3\frac{1}{5} - 2\frac{2}{3}$
	G convert a fraction to a decimal or a percentage	$\frac{3}{5} = 0.6 = 60\%$ $\frac{4}{9} = 0.4444... = 44.4...\%$
	H understand and use unit fractions as multiplicative inverses	$3 \div 5 = 3 \times \frac{1}{5}$

	Students should be taught to:	Notes
1.3 Decimals	A use decimal notation	
	B understand place value	
	C order decimals	
	D convert a decimal to a fraction or a percentage	Terminating decimals only
	E recognise that a terminating decimal is a fraction	$0.65 = \frac{65}{100} = \frac{13}{20}$
1.4 Powers and roots	A identify square numbers and cube numbers	
	B calculate squares, square roots, cubes and cube roots	
	C use index notation and index laws for multiplication and division of positive and negative integer powers including zero	
	D express integers as a product of powers of prime factors	$720 = 2^4 \times 3^2 \times 5$
	E find highest common factors (HCF) and lowest common multiples (LCM)	
1.5 Set language and notation	A understand the definition of a set	
	B use the set notation $\cup$, $\cap$ and $\in$ and $\notin$	$\mathcal{U}$ = universal set $\emptyset$ = empty set
	C understand the concept of the universal set and the empty set and the symbols for these sets	
	D understand and use the complement of a set	Use the notation A'
	E use Venn diagrams to represent sets	

	Students should be taught to:	Notes
1.6 Percentages	A understand that 'percentage' means 'number of parts per 100'	
	B express a given number as a percentage of another number	
	C express a percentage as a fraction and as a decimal	
	D understand the multiplicative nature of percentages as operators	$15\% \text{ of } 120 = \frac{15}{100} \times 120$
	E solve simple percentage problems, including percentage increase and decrease	
	F use reverse percentages	In a sale, prices were reduced by 30%. The sale price of an item was £17.50 Calculate the original price of the item
	G use compound interest and depreciation	
1.7 Ratio and proportion	A use ratio notation, including reduction to its simplest form and its various links to fraction notation	Express in the form $1 : n$
	B divide a quantity in a given ratio or ratios	Share £416 in the ratio 5 : 3 or 4 : 3 : 1
	C use the process of proportionality to evaluate unknown quantities	
	D calculate an unknown quantity from quantities that vary in direct proportion	s varies directly as t Find the missing value in a table
	E solve word problems about ratio and proportion	Including maps and scale diagrams
1.8 Degree of accuracy	A round integers to a given power of 10	
	B round to a given number of significant figures or decimal places	
	C identify upper and lower bounds where values are given to a degree of accuracy	
	D use estimation to evaluate approximations to numerical calculations	By rounding values to 1 significant figure
1.9 Standard form	A calculate with and interpret numbers in the form $a \times 10^n$ where n is an integer and $1 \leq a < 10$	$150\,000\,000 = 1.5 \times 10^8$

	Students should be taught to:	Notes
1.10 Applying number	A use and apply number in everyday personal, domestic or community life	
	B carry out calculations using standard units of mass, length, area, volume and capacity	Metric units only
	C understand and carry out calculations using time, and carry out calculations using money, including converting between currencies	
1.11 Electronic calculators	A use a scientific electronic calculator to determine numerical results	